Project Stage 1

Assignment Group 17

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1. Individual Component – xiwu3008

1. Topic and research question

The *Public Health (Tobacco and Other Products) Act 2023* issued by the Australian government came into effect on April 1, 2025, and one of the important measures is to require health message to be printed on cigarette filter to help people understand the dangers of smoking. Moreover, the government website mentions that an active smokers have a 30 to 40% higher risk of developing diabetes than non-smokers. Following this, I will explore and analyze whether there is a causal relationship between smoking and diabetes. Since most countries prohibit smoking among adolescents, there is a potential link between smoking and age, and the relationship between smoking and diabetes may not be significant in younger age groups. Therefore, the first research question I posed is: **Does smoking increase the risk of diabetes at different age groups?** Research based on this question has many stakeholders, including prediabetics, diabetics, smokers, tobacco manufacturers, retailers and others. People with prediabetes and diabetes can learn more about diabetes and the relationship between smoking and diabetes through research, so that they can improve their lifestyle away from cigarettes and increase their chances of recovery. Smokers can also learn about the dangers of cigarettes from the study, so as to reduce the frequency of smoking or quit smoking, improve prevention awareness. However, tobacco manufacturers and retailers, or other groups engaged in tobacco-related businesses, may lose revenue and profits because of the research.

Body Mass Index (BMI) is generally used to classify an individual into different weight category, and the higher the BMI, the deeper the degree of obesity. There is a consensus that obesity leads to an increased risk of many diseases, in light of this, I would like to investigate whether there is a potential relationship between BMI and hypertension and hypercholesterolemia. Because BMI classification criteria are different for adults and adolescents, and high blood pressure and high cholesterol are more common in adults, the study will focus on adults. Hence the second research question is: **Does a high BMI mean a greater risk of high blood pressure and high cholesterol in adults?** Stakeholders in the research include people with high blood pressure or high cholesterol, doctors, fitness coaches, dietitians, and people working in the fast-food. Patients with high blood pressure and high cholesterol can understand the impact of BMI on disease, enhance weight management awareness, and increase the chance of cure. Doctors can understand the relationship between BMI and disease and use BMI as a suggested factor to make more effective treatment and prevention plans for patients. Fitness coaches and dietitians can understand the harm of high BMI through this study, increase expertise, and develop personalized fitness or nutrition plans for customers to help them better prevent diseases, while also increasing income. However, because unhealthy eating habits often lead to a high BMI and thus an increased risk of disease, this may lead consumers to choose healthier diets, adversely affecting those engaged in the fast-food industry and reducing their income.

2. Data description

The dataset consists of 264,802 instances and 23 attributes, containing information on chronic diseases such as diabetes, high blood pressure, high cholesterol, and related important risk factors such as age, smoking status, BMI, and so on. The first attribute is the index of the dataset and has no special meaning. The other 22 columns described the basic information and health status of respondents. The Smoker, HeartDiseaseorAttack, Sex, BloodPressure, CholCheck, Stroke, PhysActivity, Fruits, Veggies, AnyHealthcare, DiffWalk, Cholesterol, Alcoholic and NoDocbcCost are 14 binary nominal attributes, while the 5 attributes of BMI, Physical (days), Age, Income and Mental (days) are ratio data. And the remaining three attributes, GeneralHealth, Education and Diabetes are ordinal data.

Metadata:

- Title: Diabetes Health Indicators Dataset
- Date of Creation: August 2016
- Creator/Author: Centers for Disease Control and Prevention (CDC)
- Contributor: Alex Teboul updated in November 2021
- License: CC0: Public Domain
- Format: CSV with comma as delimiter
- Data Source: The initial dataset was released on the official website of Behavioral Risk Factor Surveillance System (BRFSS), run by CDC in August 2016, then cleaned and consolidated by Data Scientist Alex Teboul and published on the Kaggle platform in November 2021. The current version was published on Canvas by the COMP5310 teaching team on March 13, 2025 as a dataset for the project.

- Collection Methodology: The data was collected by interviews conducted in 53 United States states in 2015 by the BRFSS, through Computer-Assisted Telephone Interview systems.
- Data Quality: The dataset contains many missing and invalid values and lacks completeness and accuracy.
- Size: 23MB

3. Data quality and cleaning

3.1 Data ingestion

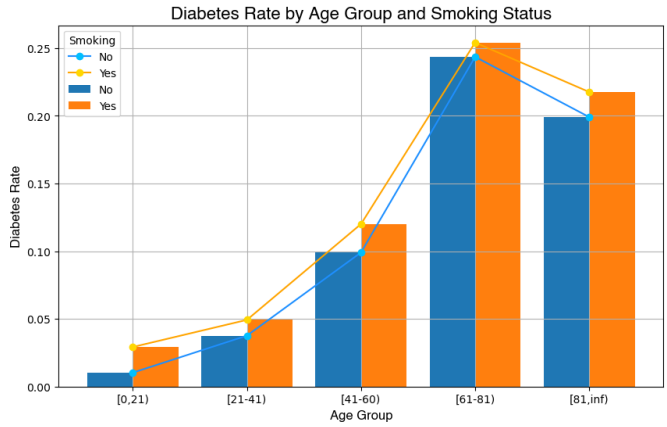
The data is read directly into the dataframe format using the read_csv method built-in the Pandas library, with the first row specified as the column name. And the data is stored in a two-dimensional dataframe data structure, with rows of 264,802 instances and columns of 23 attributes stored as Series objects that is a one-dimensional array and can be accessed by column names. At the moment of loading, the data is assigned to the appropriate Pandas data types, including int64, float64, and object representing integer, float, and string data respectively.

3.2 Data quality assurance and cleaning

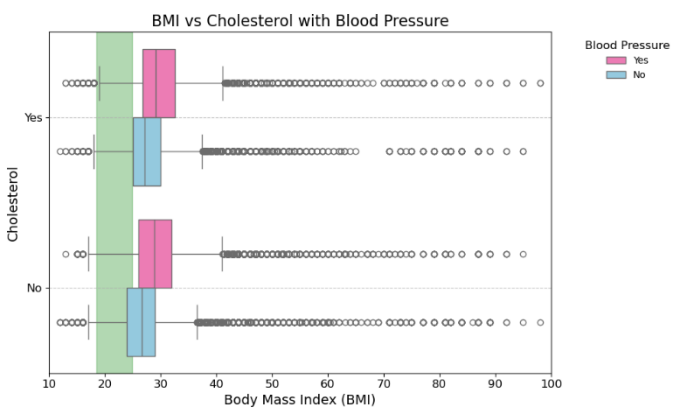
1. The value of ***Unnamed: 0*** is unique and non-null, increasing by 1 from 0, so it is considered as the index of the dataset and is renamed to 'Id' by *rename* function in Pandas. All columns with more than 50% missing values are removed because they cannot provide enough information for analysis.
2. Rows with the missing values in ***Diabetes*** are deleted since it is the target variable of study, filling them may lead to inaccurate result. And the number of missing values is very small, so removing them will not make a significant impact to the data. Using matplotlib's boxplot function, it found that some BMIs have values below 0 and above 100.
3. Using boxplot function of Matplotlib, ***BMI*** values below 0 and above 100 are identified as anomalies. Values below 0 are impossible, and those above 100 are highly unusual in practice, potentially introducing bias into the research. Since these anomalies account for only 3.34% of the data, removing them has minimal impact, so the corresponding rows are deleted.
4. The ***Smokers*** variable should contain only 1 (yes) and 0 (no), any other non-null values are considered invalid and removed to prevent errors, and these values make up just 2.74% of the data, their removal have no significant effect on overall data. In addition, the semantics of the column are transformed, using Yes and No instead of 1 and 0 to increase data readability and improve data quality.
5. ***Mental (days)*** is an interval from 0 to 30, and all entries outside this range are deleted to increase data reliability. Because the invalid values account for only 1.27% of the total, deleting them will not have a large impact on the overall data.
6. ***Age*** describes the age of the respondent, so all values below 0 or above 150 are considered invalid and deleted to ensure the validity and quality of the data.
7. ***Income*** is a string containing currency symbols and commas in the original dataset. To facilitate data processing and analysis, the currency symbols and commas are deleted by *replace* and *lstrip* functions. At the same time, since it has very few null values, the null value is removed and then converted it to an int type through the *astype* function.
8. Calling the *random.choice* function in the NumPy library to construct a *lambda* function to randomly extract values from the specified data pool according to the specified proportion, and then calling the *apply* function to apply the *lambda* function to the attributes of ***Smoker, HeartDiseaseorAttack, GeneralHealth, Sex, Education, BloodPressure, Cholesterol*** and ***Alcoholic***. By this way, the missing values of the categorical data are randomly filled based on the proportion of the data. This filling method is simple and ensures the distribution of data to the greatest extent, thus improving data quality. However, in cases where the data categories are highly unbalanced, the filling of the categories with a small proportion may be inaccurate. And the method does not take into account the relationship with other features, such as categories with different distributions under different feature combinations, which may introduce noise.
9. Finally, the missing values in numeric columns of ***BMI***, ***Mental (days)*** and ***Age*** are filled by calling the *IterativeImputer* function in the fancyimpute library and the *RandomForestRegressor* function in the sklearn.ensemble library. This method creates an iterative imputer to iteratively execute RandomForestRegressor on datasets, specifying a maximum of 5 iterations via *max_iter=5* to improve efficiency. Moreover, *random_state=42* was set to ensure the data reproducible. By comparing mean filling and median filling by kernel density plot, it can be found that iterative random forest filling is the best to maintain data distribution and improve data quality.
10. In addition to the functions mentioned above, *isna*, *notna*, *sum*, *count*, *dropna* and *shape* functions are used to view null and non-null values, find sum or counts, drop missing values and find dimension of dataframe.

4. Exploratory Data Analysis

In order to explore some trends and associations in the data, Matplotlib was used to plot a bar graph of **Diabetes Rate by Age group and Smoking Status**. At the same time, line charts are overlaid on the graph to enhance the representation of trends. It can be clearly seen from the figure that the diabetes rate increases with age and reaches its peak after age 60, which reveals the increasing trend of diabetes variables in age feature. The smoke status are emphasized by colors, and the figure illustrates that smokers have higher diabetes rates than non-smokers in all age groups. Despite Food and Drug Administration (FDA) prohibits the sale of cigarettes to adolescents under the age of 21, the number of smokers in this age group is still high. And the difference in diabetes rates is more significant among smoking status in the first age group [0, 21), which is likely to superimpose the effect of age, since within this group, smokers are mostly concentrated in older age. From the above observations can be drawn the answer to the first research question: Yes, smoking increases the risk of diabetes in different age groups.



To explore the effect of age on smoking status, boxplot and statistical descriptions of variance inflation factor (VIF) and Spearman coefficients were used. From the boxplot [Figure I Age Distribution by Smoker], smokers are more concentrated in the older group, but the difference is very slight. There are only two variables here, so the VIF of Age and Smoker is the same 1.0045, which is too small to show multicollinearity between the two features. Spearman coefficient is 0.071, with a p-value of less than 0.05, which is a tricky result, considering the previous conclusions, it can infer that although the result is statistically significant, the weak correlation near zero indicates that age and smoking status can be considered independent of each other. Based on the above conclusions, it can be found that age and smoke status are two features that have an important impact on the variables of diabetes, and these two features are independent of each other. Based on this result, it is conducive to modeling to predict the outcome of diabetes in the next stage.



Moreover, the effects of BMI on cholesterol and blood pressure in adults were also explored by left boxplot. People with high cholesterol have a slightly higher distribution of BMI than people without high cholesterol, and this mild pattern can be observed even at different blood pressure levels. Similarly, people with high blood pressure have a significantly higher BMI distribution than people without high blood pressure, and this trend is relatively apparent in both the high cholesterol and non-high cholesterol groups. In light of this, the answer to the second research question is revealed: Yes, a high BMI is associated with a greater risk of high blood pressure and high cholesterol in adults.

In addition, VIF and Spearman coefficients were used to further investigate the relationship between the three variables. The VIF of BMI, Cholesterol, and BloodPressure are 1.0497, 1.0499, and 1.0849 respectively, which suggested that there was no conspicuous multicollinearity between the three variables and that the parameters of these variables as individual predictors are not significantly affected. The Spearman coefficient between BMI and BloodPressure is 0.1345, the Spearman coefficient between Cholesterol and BloodPressure is 0.2082, and the Spearman coefficient between BMI and BloodPressure is 0.2472. The p-value corresponding to all coefficients is much less than 0.05. Considering the medical context, 0.1345 is not considered a weak correlation, while a Spearman coefficient above 0.2 is a larger correlation. Combined with the statistically significant p-value and the conclusions of the boxplot, it can be highlighted that BMI, BloodPressure, and Cholesterol pairwise are related. This conclusion is disadvantageous for the next stage of modeling, and many models such as multilinearity require features to be independent of each other, which contradicts this conclusion.

Another import observation is that the vast majority of respondents have higher BMI than Normal level which are marked by green areas in the graph, which may imply that the data is biased.

2. Individual Component - yzan0977

1. Topic and research question

The dataset that I chose is “auto_price.csv”. This is focused on the different car brands type and price etc. The topic of this dataset is vehicle pricing and influencing factors. My research question is that **will mileage, vehicle configuration and performance, as well as the age of use, have an impact on the price of a car?** The stakeholders are consumers and second-hand car dealers. As a result, the predict model trained by this dataset can help consumers and second-hand car dealers estimate the car price in the car market and achieve a fair price of a certain car. They won't get ripped off because of this dataset.

2. Data description

In the dataset, there are 23 attributes (not including the serial numbers in the first column) and 18286 instances. The data dictionary will be included as an appendix and then I will give the meta data (Duygujones, n.d) below.

Metadata:

- **Title:** AutoScout24_Car_Sales2019
- **Data of Creation:** June 2024
- **Collaborators:** Duygu Jones (Owner)
- **License:** Apache 2.0
- **Format:** CSV
- **Data Source:** Initially from https://www.autoscout24.com, then uploaded to the website Kaggle. The current version was published by the COMP5310 teaching team and can be downloaded from Canvas since Mar 13, 2025 8:00 until Apr 8, 2025 23:59.
- **Data Quality:** Containing missing values and abnormal values
- **Size:** 7306 KB

3. Data ingestion and cleaning

3.1. Data ingestion

I use Pandas and Python to read the csv file. The data structure is DataFrame. DataFrame is one of the most commonly used and powerful data structures in Pandas libraries, and it can be thought of as a two-dimensional tabular data structure, similar to a table in a spreadsheet or a relational database.

3.2. Data quality assurance and cleaning

When clearing the data, in the Jupyter I first load “auto_price.csv”. The DataFrame is defined as **df**. Then I use the functions **df.info()**, **df.head()** and **df.describe()** to preview the basic condition of the data. Then I divide the number of null values by the total number to find the proportion of missing values and print it. From the outcome, we can see that for the three attributes: "Extras","Safety_Security" and "Upholstery_Type", the missing rates are all over 70%. And for “Paint_Type”, I believe this attribute is unlikely to affect the price of the car, because customers can change the car to any color with the film. As a result, I dropped these four columns. Meanwhile, I didn’t find a duplicate value by using **df.duplicated().sum().** After that we come to the process of handling inconsistent data types and formats.

Through careful screening and observation of the data, we can find that in the column “Weight”, it has multiple units: “g”, “kg” and “lbs”; in the column “Price”, it has multiple units: “$”, “€” and “£”; in the column “Mileage”, it also has multiple units: “km” and “mi”. Then I define three functions to handle these situations: to do currency exchange and unit standardization. Also, I do some little adjustment towards the “Displacement”: change the “string” to “float”; for “Horsepower”: change the unit KW to Horsepower and convert some string list to Json format. Then, I use the IQR function to help me remove abnormal values. After the correcting process, I will introduce my methods of imputing missing values.

By logical thinking and common sense of life (AutoScout24. n.d), we can know that for specific Make_models and Body_types, its factory weight and displacement are only a few standard values. As a result, we can fill in the missing value of “Weight” and “Displacement” using the mode from the known “Make_Model” and “Body_Types”; at the same time, we can

inversely fill the rows with a specific “Displacement” and “Weight” that have missing values in the “Make_Model” and “Body_Types”, then I will drop those rows that still have missing value in “Make_Model” and “Body_Types”, which means the value of that row may be not reliable. And I defined the function **dealwith_mm_and_bt()** to implement this target. After that, the remaining missing value of other attributes is very easy to impute grouped by “Make_Model” and “Body_Types”. I use mean value to fill missing value interval, ratio and usage-based attributes such as “Price” and “Mileage”, then use mode value to fill nominal, ordinal and technical attributes such as “Fuel” and “Gears”. Generally, I define a bunch of functions to clean the data, all of them are named like **“dealwith_xxxx()”** in the Jupyter code file, then I define the **clean_data()** function to run them all.

After all the above methods, all missing values are imputed, and all abnormal values are removed, the cleaned dataset has 14986 rows.

4. Exploratory data analysis (EDA)

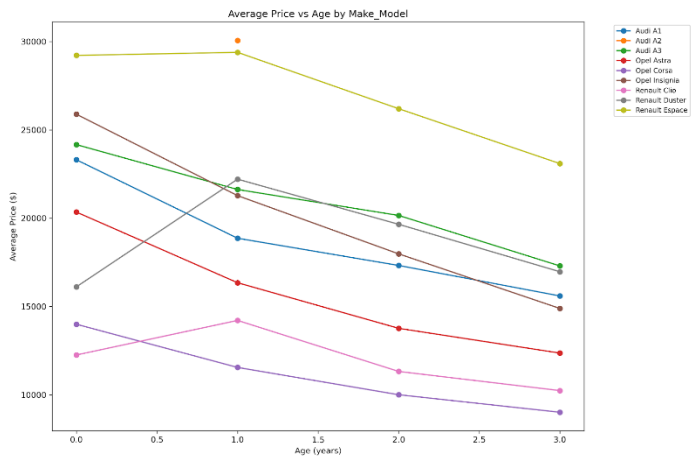


Fig. 4.1.

According to common sense in real life, the number of “Mileage” is proportional to the “Age”, and maybe the “Price” will go down as the “Age” goes by. So, I use pyplot to draw line plots ***Fig. 4.1.*** to explore if this is true. Surprisingly, while for most “Make_Model”s its price does decline with “Age”, for Renault it is even more expensive in “Age” one than “Age” zero. This may be caused by a number of factors, and it is worth exploring further. The effect of this result on subsequent modelling results is that in the resulting model, the price of a particular vehicle is not always inversely proportional to the age of the car.

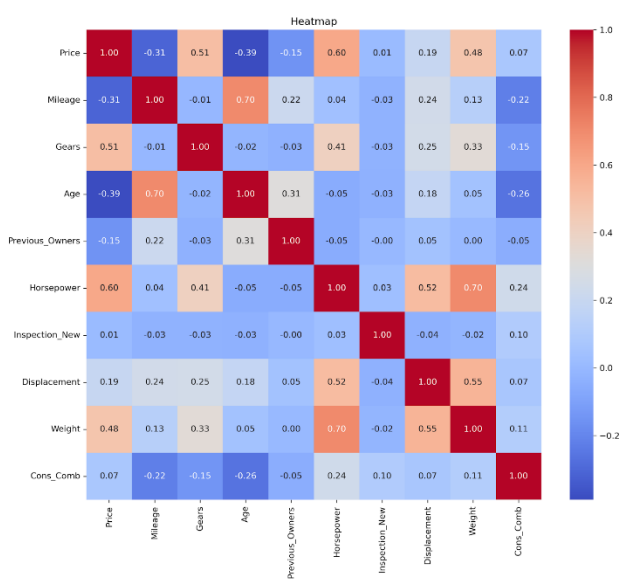


Fig. 4.2.

Because I wanted to learn more about which numerical properties might be correlated except “Price” and “Age”, I used a heat map ***Fig. 4.2.*** to show all the numerical attributes in the dataset. What I didn't expect is that the “Gears”, “Horsepower” and “Price” are also related. Whether the high price is due to technical reasons, or because people prefer more powerful cars, I also need to further model and carefully study in the future. The effect of this result on the subsequent modelling results is that it provides some guidance on the linear relationships between properties that may be predicted by subsequent models.

3. Individual Component - kwan0569

1. Topic and research question

The dataset I selected is "forest_cover". I think this topic is a study of forest type classification of topographic and environmental features. Because the data includes topographic features, environmental factors, soil types, forest types so on and so forth, there are two questions that I ask. The first question is which characteristics are most representative in determining forest types? Which is more important, Elevation, Soil Type or Hillshade? Second, how do elevations affect the distribution of different forest types? Whether certain forest types are concentrated at specific elevations to provide forest conservation and planning methods for forest resource managers or forest service staff. Finally, I hope that this dataset will be useful to the forestry sector, for example, it can help forestry workers to plan forest areas and understand the different types of tree species suitable for planting at different elevations.

2. Data description

For metadata, I queried four sites in the dataset, Neota, Rawah, Comanche Peak and Cache la Poudre. I found the locations in Roosevelt National Forest in northern Colorado, and Google traced the data to Covertype. The data was donated on 7/31/1998(UCI Machine Learning Repository.1998). The dataset has 30,860 rows and 56 columns. There is data missing in many columns, and the missing ratio of some columns is high, especially the “Hillshade” and “Soil_Type” columns. The target variable in this dataset is “Forest_Cover”, which means the forest cover type label, such as "Lodgepole Pine", "Ponderosa Pine", "Aspen", etc. Terrain and environmental features include “Elevation”, “Aspect”, “Slope”, “Hillshade_9am/Noon/3pm”, “Horizontal_Distance_To_Hydrology”, “Vertical_Distance_To_Hydrology”, “Horizontal_Distance_To_Roadways”, “Horizontal_Distance_To_Fire_Points”.

Metadata:

- **Title**: Covertype
- **Date of Creation**: July 1998
- **Creator/Author**: UCI Machine Learning Repository.
- **License**: a Creative Commons Attribution 4.0 International (CC BY 4.0) license.
- **Format**: CSV
- **Data Source**: The actual forest cover type for a given observation (30 x 30 meter cell) was determined from US Forest Service (USFS) Region 2 Resource Information System (RIS) data. Independent variables were derived from data originally obtained from US Geological Survey (USGS) and USFS data. The current version was published on Canvas by the COMP5310 teaching team on March 13, 2025 as a dataset for the project.
- **Data Quality**: There are a large number of missing and abnormal values in the dataset, so the data is not accurate and not very representative
- **Size**: 10.7MB

3. Data ingestion and cleaning

3.1. Data ingestion

I used pandas python to obtain the data, and the data structure was in DataFrame format. Dataframes are a popular abstraction to represent, prepare, and analyze data (Petersohn et al., 2020). And it is one of the most core data structures in the Pandas. I often use the dataframe format for operations such as reading csv files and viewing file structure information.

3.2. Data quality assurance and cleaning

When I clean data, first load my dataset "forest_cover.csv" to view the basic information of the data, and then I use the “df.drop()” function to remove the index column to keep useful information in the dataset. Next use “df.isnull()” function to look at the number of missing values for each field. Because "Soil_Type" is in one-hot encoding format, in one-hot encoding, only one “Soil_Type” should be listed as “1”, and the other should be “0”. So if missing values exist, using “fillna(0)” function to add with “0” keeps the logic intact, and adding with “0” is less likely to cause an error than adding with “1”. Next I use “df.select_dtypes()” function to select all columns with value types “int64” and “float64”, and exclude “Soil_Type” because so far there is no format for unifying numeric types. I then use the IQR method to detect and replace the abnormal value with “Nan”. This can try my best to decrease the error caused by missing values.

Then I looked up a class on BiliBili to learn the difference between using the mean, median, mode to populate data, and using the median to populate Nan values. Because using the average may make the results too high and inaccurate. While the median will have a small error, try to avoid the interference of abnormal value. However, the mode representing "the most occurring value" is not suitable for filling this dataset, because the data of each column is based on the data collected in reality, such as elevation, slope, etc., and it is difficult to achieve the same data. So, I filled it with the median by “fillna(median_val)” function.

After that I convert all float data to int, and for the sake of subsequent visualizations, I convert the “Forest_Cover” column into a class variable and generate the corresponding encoded column. Then I check and remove duplicate values. Finally, I found that there were many irrational values in the data, such as negative values on “Slope” and “Hillshade” beyond the reasonable range (255). Because “Hillshade” is usually represented as an 8-bit grayscale image, the brightness value of the grayscale image ranges from 0 to 255, that means, fully shaded - fully illuminated. So, I specify a range for each attribute and use a mask to clean values that don't fall within that range. In Pandas, the mask is an array of True/False that indicates whether each row is "reserved" or not. Finally, I saved the cleaned dataset as a “cleaned_forest_cover.csv” file. So, I can then use this cleansed file for Exploratory data analysis (EDA).

4. Exploratory data analysis (EDA)

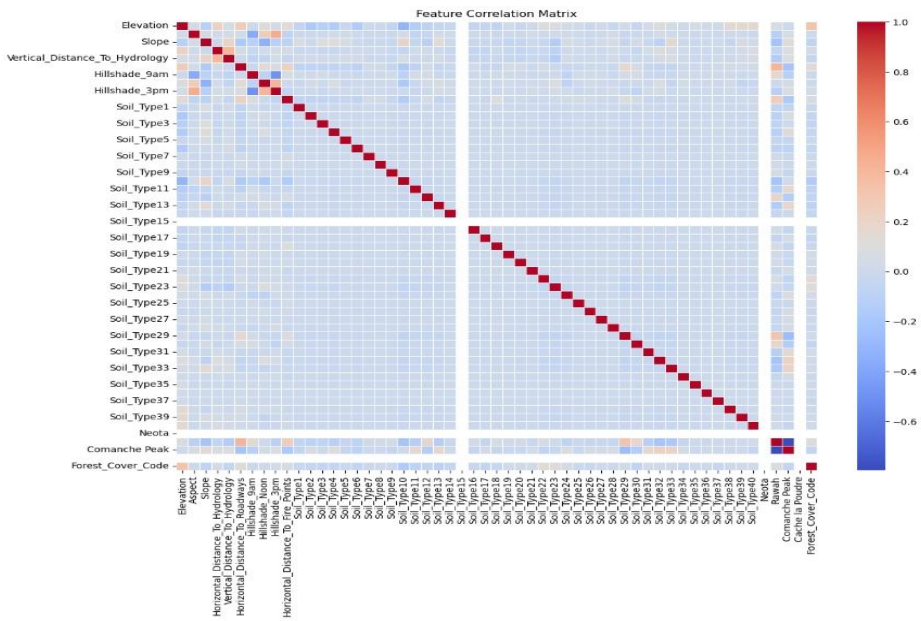


Fig. 4.1.

First, because the purpose of this dataset is to determine the type of forest in a place based on topography, soil and environmental characteristics, I drew Fig. 4.1 with Pearson Correlation coefficient, which can reflect the "linear relationship". In this figure, the color represents the strength and direction of correlation, that is, red (+1): strong positive correlation, blue (-1): strong negative correlation, and white is neutral relationship. Because the diagonal is all red, it means that each variable is completely related to itself, so there is no information. The darkest features in this figure include Elevation, some Soil_TypeX columns, Comanche Peak, Neota and other regional variables, indicating a strong relationship between Elevation and forest type, possibly because elevation in different regions affects forest type distribution. “Soil_Type” may strongly affect forest type distribution; Neota and Comanche Peak have significant effects on “Forest_Cover_Code”. Moreover, there is no obvious collinearity between most numerical variables, so there is no need to worry too much about feature redundancy. The most important attributes are clearly identified in stage 2 modeling.

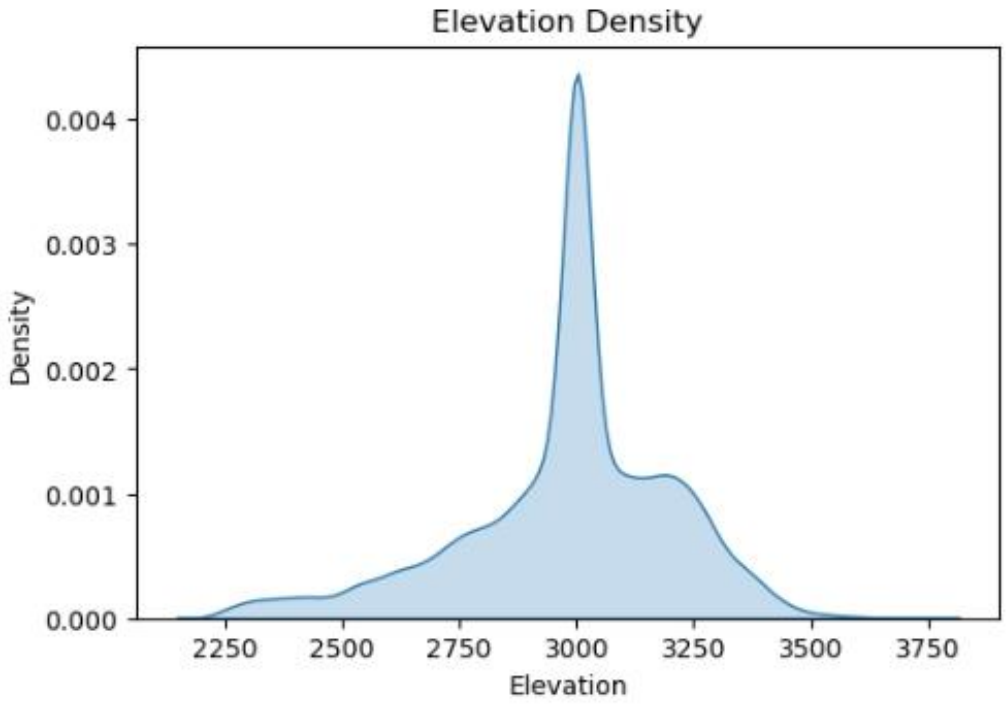


Fig. 4.2.

From Figure 4.1, I can see that “elevation” is an important factor affecting the distribution of forest types, so I made a Descriptive Statistics of elevation attributes. Among them, the mean of elevation attribute is 2983.05, the median is 3002, the standard deviation is 224.52, the IQR is 232.0, the maximum value is 3733, the minimum value is 2231, the data type is int, and the variable type is ratio. Therefore, I drew Fig.4.2: Kernel Density Estimate (KDE) of elevation, to show the elevation distribution pattern in the dataset.

From Figure 4.2, I can see that the abscissa represents elevation and ranges from about 2200 to 3750 meters, and the ordinate represents density values rather than "frequency" because I need the probability density to tell where elevation occurs more frequently (the sum of areas under the curve is 1). I can find that the elevation of a large number of samples in the dataset is concentrated between about 2950-3050 meters, which is the "core area" of the data. The peak value is very sharp, indicating that the elevation distribution is very concentrated, and the standard deviation is small. There is also a small peak at about 3,200 meters; elevation ranges above 3000 m have slightly more values than lower elevations, and the overall distribution is skewed to the right and therefore not a standard normal distribution. In the modelling of stage 2, modelling methods related to normal distribution cannot be used, which may lead to inaccurate models.

4. Discussion

4.1 Discussion on diabetes_diagnosis.csv

Discussion for Dataset

Strengths: The data size is large enough to investigate for research questions; Variable types are rich, including interval, ordinal, nominal, and ratio, which are helpful to analyze the relationship between different types of variables, such as exploring the relationship between BMI and BloodPressure in research questions; The data background is clear, it is real data obtained by BRFSS interview, which makes it easy to understand and interpret, especially when interpret the result of research questions; Attributes are not completely independent of each other, which makes it easy to find trends in data for research question, but is not conducive to the next stage of modeling.

Limitations: There are biases in the data, such as most BMIs exceeding the normal level, which may introduce bias in the for investigation on relationships between BMI and cholesterol and blood pressure in the second research question; There are too many missing values and random filling may cause data bias, especially for those categories with different distributions under different feature combinations, such as in the research of first question, explore the relationship between age and smoke status, which may be influenced by bias and lead to wrong conclusions.

Discussion for EDA

Strength: Combined with appropriate figures and statistical descriptions, the relationship between features is explored in many aspects, and which make the result more comprehensive. Identified the skewness in BMI variable, and revealed some trends in the data.

Limitation: VIF and Spearman coefficients are more suitable for continuous or ordinal data, so it may not fully capture the relationship between nominal and numerical variables. It mainly discusses the relationship between variables, and lacks the discussion on the distribution of a single variable, resulting in a single research direction.

4.2 Discussion on auto_price.csv

Discussion for Dataset

Strength: For the dataset “auto_price.csv”, its advantage is that the correlation of various attributes is relatively strong, and the correlation of some attributes can be inferred through common sense of life (such as price and ages, mileage, etc.), and the missing value is also easy to fill because of this reason.

Limitation: The disadvantage is that the amount of data is not as large as the other two data sets, only some models are involved, the future model prediction may be inaccurate, and there are many string-type attributes in car configuration, which need to be further processed during modelling, and the data consistency and standardization problems are serious, too. For the questions **yzan0977** wants to explore, the advantages of this data set can help the model predict the car price more easily, while the disadvantages will cause some perturbations to the model prediction.

Discussion for EDA

Strength: For EDA, the advantage of using line charts to show the relationship between prices and years is that the growth trend of prices of different make_models with ages can be clearly displayed, and multi-series comparisons can also be made through make_mode and body_type attributes.

Limitation: The disadvantage is that aggregated data can affect the dispersion and fluctuation of the original data. For the use of heatmap, the advantage is that it can find the linear correlation between the car price and the numerical data, but the disadvantage is that it will ignore the nonlinear relationship and non-numerical data. For the questions **yzan0977** wants to explore, the advantage of using line charts and heat maps is that when the model predicts the car price, it will pay more attention to the linear relationship of the attributes, and the disadvantage is that the nonlinear relationship between the attributes and the car price will be ignored in the prediction.

4.3 Discussion on forest_cover.csv

Discussion for Dataset

Strength: I think the dataset “forest_cover.csv” has the following four advantages: First, there are many attributes in this dataset, including elevation, slope, hillshade, soil type, the distance from water sources, road and fire points, and regional distribution. Through the above attributes, I can understand that the forest cover rate is related to these factors. Moreover, by studying geography I learned that elevation, soil type and hillshade are the most representative characteristics, because these three data are the main factors affecting forest distribution. And elevation is the most important factor. Because different elevations affect temperature, different temperatures determine the formation of local forest types. Second, the sample size in this dataset is so large, with 30,000 + samples, which is beneficial to reduce errors in stage1 visualization and stage 2 modelling. Third, the variables of this dataset are clear, for example, Forest_Cover has 7 different tree species, which is suitable for classification visualization. Visualization via scatter plot shows that Pomderosa Pine is distributed at 2200-2500 meters above sea level, Lodgepole Pine is mainly distributed at 2600-3000 meters above sea level, and Spruce is widely distributed at 3000-3300 meters above sea level. However, there are a few Spruce above 3400 meters, and finally Krummholz is mostly between 3400 and 3800 meters above sea level. By looking at the distribution of these different species at different elevations, it is believed that forestry workers and local forest authorities can clarify how to plan forests so that trees are more survivable and least affected by the elements. Finally, I can use this dataset for a variety of modelling at stage 2, not only to classify forest types, but also for feature selection or anomaly detection.

Limitation: But the disadvantages of the dataset are obvious too. First, as soon as I open the file, I see a lot of empty cells and there are a lot of columns with missing values. This will cause a lot of trouble when I do the data cleaning and data analysis. Second, there are serious abnormal values in many columns. For example, “elevation” is negative, “Slope” is -81, and “Hillshade_9am” has a maximum value of 561, but 255 is the maximum reasonable value. Third, some attributes are not explained. For example, the meaning of “Aspect” and “Horizontal_Distance_To_Fire_Points” is not clear, which affects my do data analysis. Fourth, drawing with data that has not yet been cleaned is inaccurate. It may interfere with the work of forestry workers to some extent.

Discussion for EDA

Strength: In terms of EDA, the advantage of using heatmap is to express the strength of data through color depth, which can clearly see the close relationship between “elevation” and other columns. It is suitable for observing the relationship between multiple variables. The disadvantage of using heatmap is that if there are too many variables, the color blocks will become very dense and difficult to distinguish.

Limitation: The advantage of Descriptive Statistics analysis is that mean, median, standard deviation and distribution range can be quickly identified, and it does not rely on algorithms, it is easy and reliable, so it is suitable for early data exploration. But its disadvantages are also obvious, this method is greatly affected by abnormal values, for example, the average is easily biased by extreme values.

5. Conclusion

The data distribution and some trends among variables are revealed by EDA. For example, Elevation shows a negative skew distribution, focusing on higher values after 3200m; the diabetes rate increases with age, and smoking also increases the risk of diabetes; some particular brands can have an inverse effect on used-car prices, such as car prices of Renault have risen and then falls with the age, rather than consistent decline like other most brands. In the next stage, we will use the **diabetes_diagnosis.csv** dataset because it is large enough to reduce sample errors and be more representative, to identify details and special variations, and to better assist the model to capture features and trends, reducing the risk of overfitting.

6. Appendix

6.1 Data Dictionary

6.1.1 Data Dictionary of **diabetes_diagnosis.csv - xiwu3008**

Attributes	**Rename**	**Type**	**Description**
Unnamed: 0	Id	Int	Index of the dataset, from 0 to 264801.
CholCheck	-	float	Indicate whether the respondent has had a cholesterol check. 1.0 represents yes, and 0.0 represents no.
BMI	-	Float	Indicate body mass index of the respondent.
Smoker	-	Float	Indicate whether the respondent has smoked at least 100 cigarettes in entire life. 1.0 is yes, 0.0 is no, and other are invalid values.
Stroke	-	Float	Indicate whether the respondent has had a stroke. 1.0 is yes and 0.0 is no
HeartDiseaseorAttack	-	Float	Indicate whether the respondent has been told by a doctor, nurse or other health professional that he/she has heart disease or heart attack. 1.0 is yes and 0.0 is no.
PhysActivity	-	Float	Indicate whether the respondent has had physical activity (excluding job) in the past 30 days. 1.0 is yes and 0.0 is no.
Fruits	-	Float	Indicate whether the respondents consume fruit once or more per day. 1.0 is yes, and 0.0 is no.
Veggies	-	Float	Indicate whether the respondents consume vegetables once or more per day. 1.0 is yes, and 0.0 is no.
AnyHealthcare	-	Float	Indicate whether the respondents have any health care coverage, including health insurance, prepaid plans, government plans, or Indian Health Services. 1.0 is yes, and 0.0 is no.
NoDocbcCost	-	Float	Indicate whether the respondent have had a time in the past 12 months when he/she needed to see a doctor but could not because of cost. 1.0 is yes, and 0.0 is no.
GeneralHealth	-	String	Indicate whether the respondent considers his/her health in general to be one of the following: Excellent, Very Good, Good, Fair, Poor.
Mental (days)	-	Float	Indicate the number of days during the past 30 days, the mental health of respondent was not good. 0.0 is the minimum and 30.0 is the maximum.
Physical (days)	-	Float	Indicate the number of days during the past 30 days, the physical health of respondent was not good. 0.0 is the minimum and 30.0 is the maximum.
DiffWalk	-	Float	Indicates whether the respondent has serious difficulty walking or climbing stairs. 1.0 is yes, and 0.0 is no.
Sex	-	String	Indicate sex of respondent.
Age	-	Float	Indicate age of respondent.
Education	-	String	Indicate level of education completed of the respondent.
Income	-	String	Indicate income of respondent.
Diabetes	-	String	Indicate whether the respondent has ever been told he/she have diabetes.
Cholesterol	-	String	Indicate whether the respondent has been told by a doctor, nurse or other health professional that his/her blood cholesterol is high.
Alcoholic	-	String	Indicate whether the respondent is a heavy drinker (adult men having more than 14 drinks per week and adult women having more than 7 drinks per week).
BloodPressure	-	String	Indicate whether the respondent has been told by a doctor, nurse or other health professional that he/she has high blood pressure.

6.1.2 Data Dictionary of **auto_price.csv - yzan0977**

Attribute	Data Type (before data clean)	Description	Unit/Notes
Unnamed: 0	int	Index of csv, from 0 to 18285	-
Make_Model	string	Brand and model of the car, having 9 types	-
Body_Type	string	Body type of the car, there are 8 kinds of Body_Type	-
Price	string	Price of the car, **Not float because** contain units such as $, €, £, need to be cleaned: currency exchange	$,€,£
Vat	string	Value-added tax of the car, Price negotiable or VAT deductible	-
Mileage	string	The number of miles the car has traveled, **Not float because** contain unit km or mi, need to be cleaned: unit standardization	km or mi
Type	string	Use status of the car	-
Fuel	string	Types of energy used by the car, Benzine, Diesel, Electric or LPG/CNG	-
Gears	int	The number of gears in the car's gearbox, from 5 to 8	-
Comfort_Convenience	string	Comfort convenience properties of the car	-
Entertainment_Media	string	Entertainment Media properties of the car	-
Extras	string	Extra properties of the car	-
Safety_Security	string	Safety Security properties of the car	-
Age	int	Auto age of the car	years
Previous_Owners	int	Number of previous owners, from 0 to 4	-
Horsepower	string	Horsepower of the car, **Not float because** contains units (KW), need to be cleaned and transfer to horsepower	KW
Inspection_New	int	Whether the vehicle has passed the latest annual inspection or test, use int (1 and 0) to represent boolean value, 1 means true and 0 means false	-
Paint_Type	string	Paint Type of the car	-
Upholstery_Type	string	Upholstery Type of the car	-
Gearing_Type	string	Gearing Type of the car	-

Displacement	string	Engine displacement of the car, **Not float because** contain units(cc), need to be cleaned	cc
Weight	string	Weight of car, **Not float because** contain units(kg、g、lbs), need to be cleaned: unit standardization	kg、g、lbs
Drive_Chain	string	Driving mode of cars: front, rear or 4WD	-
Cons_Comb	float	Comprehensive fuel consumption of cars	L/100km

6.1.3 Data Dictionary of **forest_cover - kwan0569**

Attributes	Type	Measurement	Description
Unnamed: 0	int	-	Index of the dataset, from 0 to 30859.
Elevation	float64	Meters	The vertical distance of a location relative to mean sea level.
Aspect	float64	Azimuth	Aspect refers to the direction the slope faces, which can also be understood as "which side the slope faces".
Slope	float64	Degrees	Slope refers to the steepness of the terrain at a point on the surface.
Horizontal_Distance_To_Hydrology	float64	Meters	The shortest straight-line distance from a location to the nearest body of water measured head-on on a map.
Vertical_Distance_To_Hydrology	float64	Meters	The vertical difference in height between a location and the nearest body of water.
Horizontal_Distance_To_Roadways	float64	Meters	The horizontal distance between a location and the nearest road.
Hillshade_9am	float64	0 to 255 index	Depending on the terrain and the position of the sun, the light intensity of a location at 9 a.m.
Hillshade_Noon	float64	0 to 255 index	Depending on the terrain and the position of the sun, the light intensity of a location at noon.
Hillshade_3pm	float64	0 to 255 index	Depending on the terrain and the position of the sun, the light intensity of a location at 3 p.m.
Horizontal_Distance_To_Fire_Points	float64	Meters	The horizontal distance from a location to the point of fire
Soil_Type (1-40)	float64	0 or 1	Used to represent specific soil classifications for 40 sites.
Forest_Cover	float64	1 to 7	It usually indicates the type of forest cover in an area.
Neota	int64	-	An area located in the Roosevelt National Forest of northern Colorado.
Rawah	int64	-	An area located in the Roosevelt National Forest of northern Colorado.
Comanche Peak	int64	-	An area located in the Roosevelt National Forest of northern Colorado.
Cache la Poudre	int64	-	An area located in the Roosevelt National Forest of northern Colorado.

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6.3 Additional Figures

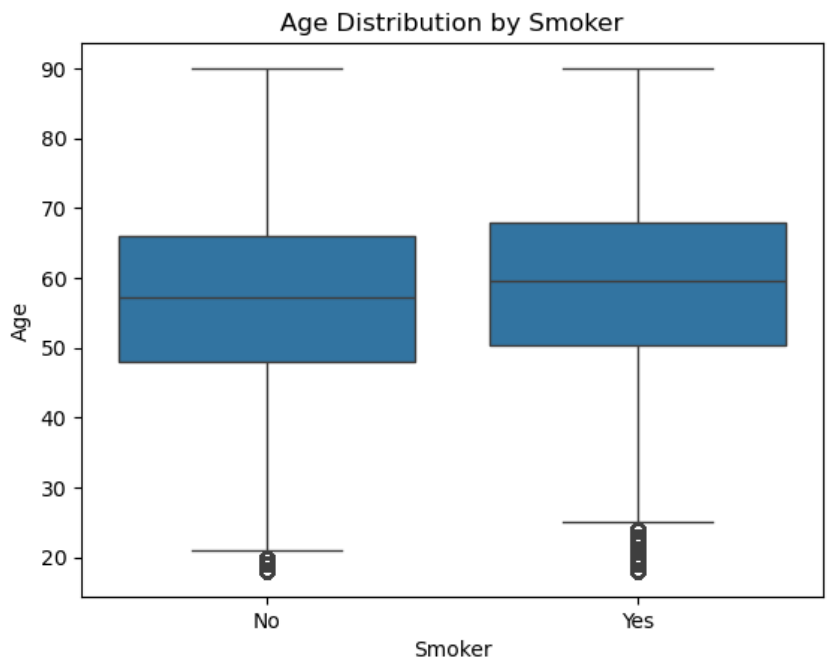

Figure I Age Distribution by Smoker

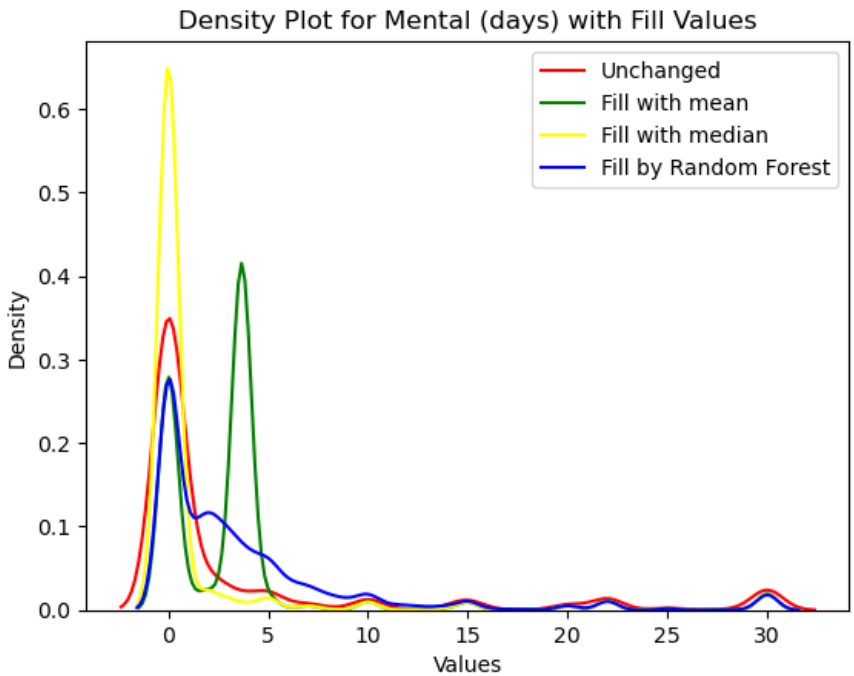

Figure II Density Plot for Mental (days) with Filling Values

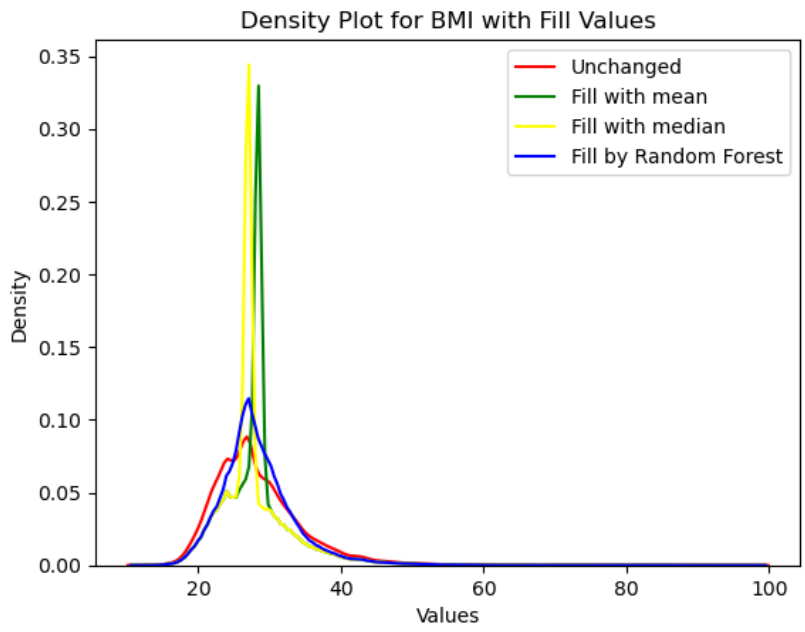

Figure III Density Plot for BMI with Filling Values

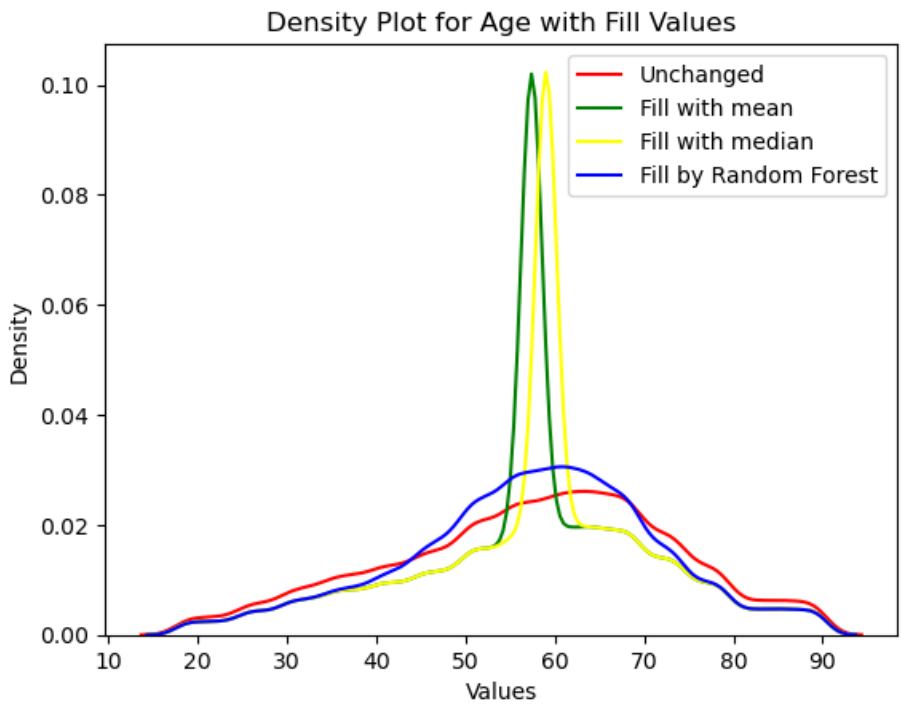

Figure IV Density Plot for Age with Filling Values